



3.2 Right-angled trigonometry and bearings

Geometry and trigonometry · IB Diploma · Concept guide and original exam-style practice

Understand the concept

In a right-angled triangle, SOH-CAH-TOA gives $\sin\theta = \text{opposite/hypotenuse}$, $\cos\theta = \text{adjacent/hypotenuse}$, $\tan\theta = \text{opposite/adjacent}$. Angles of elevation/depression are measured from the horizontal. Bearings are measured clockwise from north (000° to 360° , always given as three digits), used with trigonometry to solve navigation problems, often combined with the sine/cosine rules for non-right triangles.

Worked example: follow the reasoning

A tower is 40 m tall. From a point on level ground, the angle of elevation to the top is 35° . Find the horizontal distance to the tower.

$$\tan(35^\circ) = 40/d$$

$$d = 40/\tan(35^\circ) = 40/0.7002 = 57.1 \text{ m (3 s.f.)}$$

Quick check · multiple choice

A ship sails on a bearing of 120° for 10 km. What is the bearing to return directly to the start?

- A 060°
- B 240°
- C 300°
- D 120°

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 From the top of a cliff 85 m high, the angle of depression to a boat is 22° . [7]
(a) Find the horizontal distance from the base of the cliff to the boat.
(b) The boat then sails directly away from the cliff on a bearing of 090° for 200 m. Given the cliff base is due west of the boat's start point, find the new angle of depression from the top of the cliff to the boat, to the nearest degree.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $\tan(22^\circ) = 85/d \rightarrow d = 85/\tan(22^\circ) = 85/0.4040 = 210.4$ m.

(b) New horizontal distance = $210.4 + 200 = 410.4$ m. New angle of depression:
 $\tan\theta = 85/410.4 = 0.2071$, $\theta = \arctan(0.2071) = 11.7^\circ \approx 12^\circ$.

Quick check answer

C. The reverse bearing is $120^\circ + 180^\circ = 300^\circ$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.